Kubernetes-08

Tasks:

1. **Create and Test a Kubernetes Pod with an EmptyDir Volume**
2. **Configure a HostPath Volume in Kubernetes and Validate Data Persistence**
3. **Deploy an Amazon EBS Volume Using Persistent Volume and Persistent Volume Claim (PVC)**
4. **Set Up an Amazon EFS Volume and Attach it to Multiple Pods**
5. **Implement and Test Liveness and Readiness Probes in a Kubernetes Pod**

Task:

1.Create and Test a Kubernetes Pod with an EmptyDir Volume

We will:

1. Create Pod with EmptyDir
2. Create files
3. Restart container → verify data exists
4. Delete pod → verify data lost

Step 1: Create EmptyDir Pod

Create file:

```
vi emptydir-pod.yaml
```

Apply it:

```
kubectl apply -f emptydir-pod.yaml
```

Check:

```
kubectl get pods
```

Wait until STATUS = Running.

```
ubuntu@master:~$ kubectl apply -f emptydir-pod.yaml
pod/emptydir-pod created
[ubuntu@master:~$ kubectl get pods
NAME           READY   STATUS    RESTARTS   AGE
emptydir-pod   1/1     Running   0          6s
[ubuntu@master:~$ kubectl get pods -o wide
NAME           READY   STATUS    RESTARTS   AGE   IP               NODE      NOMINATED NODE   READINESS GATES
emptydir-pod   1/1     Running   0          14s   192.168.235.131   worker1   <none>           <none>
ubuntu@master:~$
```

Step 2: Create Test Files

Enter the pod:

```
kubectl exec -it emptydir-pod -- bash
```

Inside container:

```
cd /data

touch tiger1 tiger2 tiger3

ls
```

```
emptydir-pod   1/1     Running   0          14s   192.168.235.131
ubuntu@master:~$ kubectl exec -it emptydir-pod -- bash
[root@emptydir-pod:/# cd /data
[root@emptydir-pod:/data# touch tiger1 tiger2 tiger3
[root@emptydir-pod:/data# ls
tiger1  tiger2  tiger3
root@emptydir-pod:/data#
```

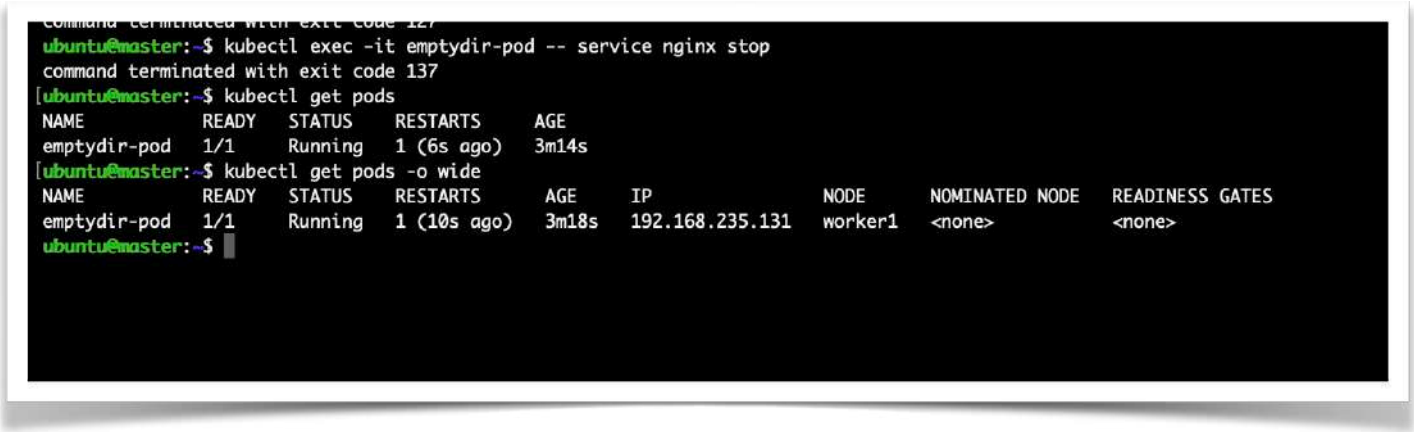
Step 3: Restart Container (Not Pod)

Now stop nginx process:

```
kubectl exec -it emptydir-pod -- service nginx stop
```

Check:

```
kubectl get pods
```



You will see:

```
RESTARTS = 1
```

Now check files again:

```
kubectl exec -it emptydir-pod -- ls /data
```

Files will still be there.

✔️ Because EmptyDir survives container restart
✔️ Pod is still same

Step 4: Delete Pod (optional just to show it does not save data if the pod is deleted)

Now delete pod:

```
kubectl delete pod emptydir-pod
```

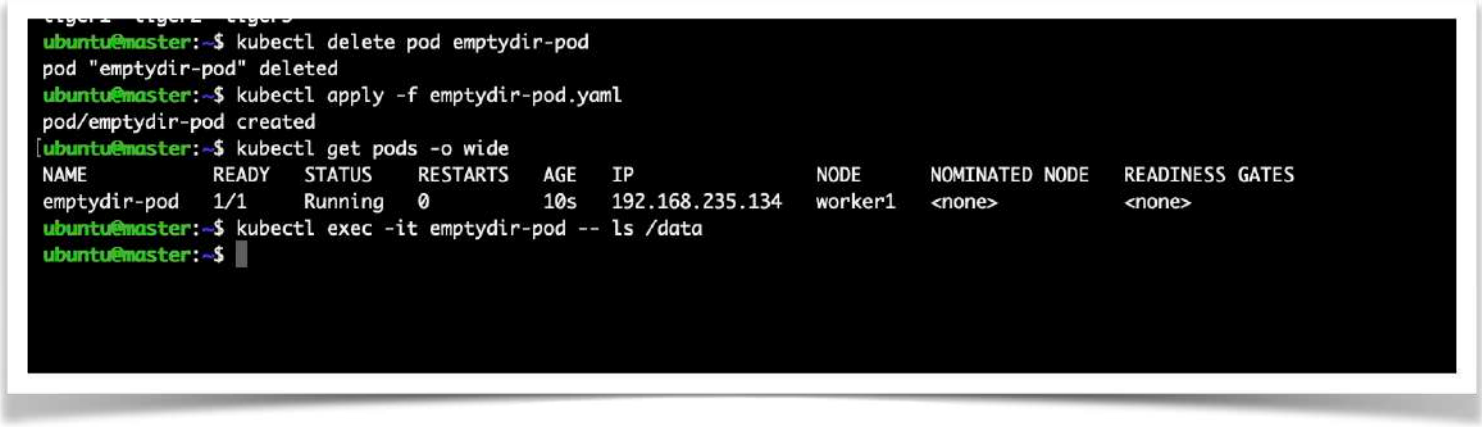
Recreate:

```
kubectl apply -f emptydir-pod.yaml
```

Now check again:

```
kubectl exec -it emptydir-pod -- ls /data
```

❌ Folder will be empty.



Final Conclusion

Action	Data Status
Container Restart	✅ Data stays
Pod Delete	❌ Data lost

Because:

EmptyDir lifetime = Pod lifetime

Task:2
2.Configure a HostPath Volume in Kubernetes and Validate Data Persistence

This one is very important for understanding **node-level storage**.

Goal

✔ Create a Pod using HostPath
✔ Create files
✔ Delete Pod
✔ Recreate Pod
✔ Verify data still exists

First Understand HostPath

- HostPath stores data on **Node machine**
- Not inside container
- Not inside Pod

- Directly on Node filesystem

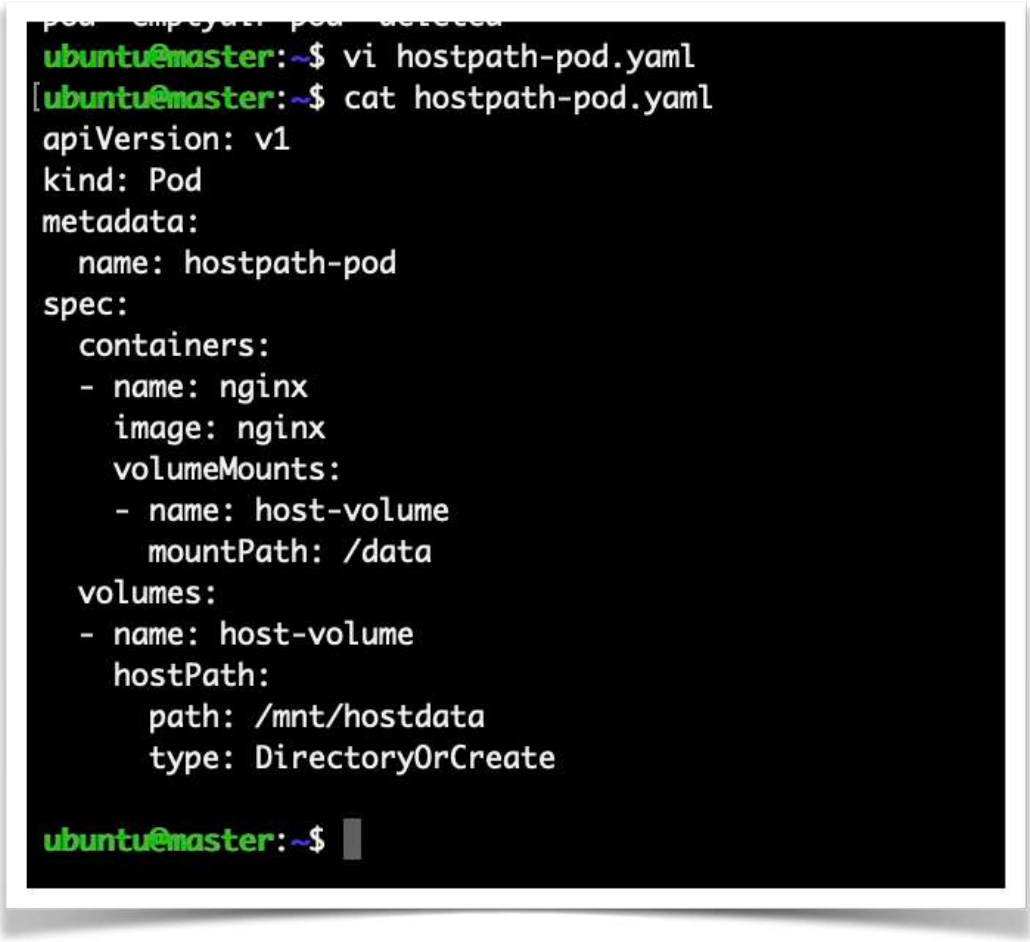
So:

Container restart → Data stays ✅
Pod delete → Data stays ✅
But if Pod moves to another node → Data NOT available ❌

STEP 1: Create HostPath Pod

Create YAML:

```
vi hostpath-pod.yaml
```



Save and exit.

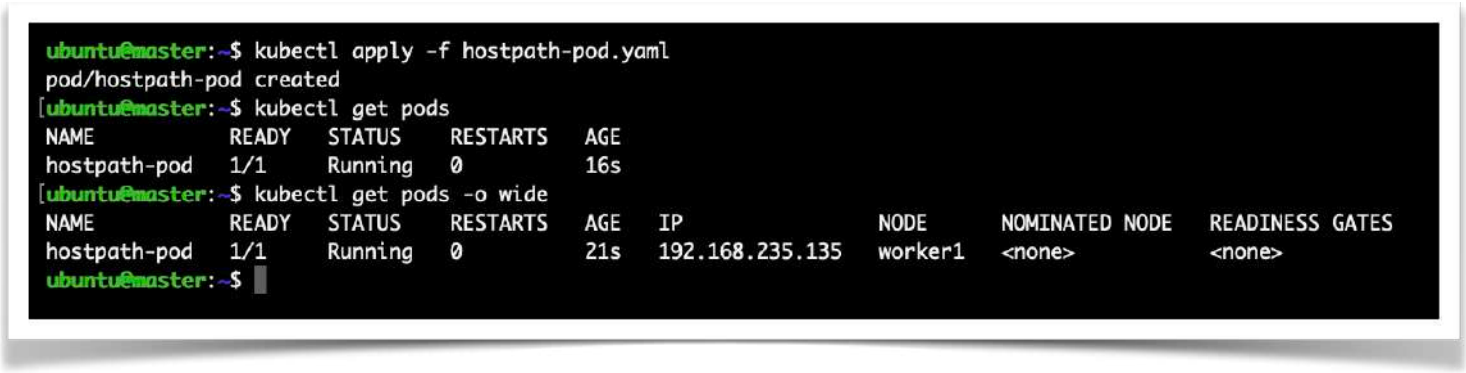
Apply it:

```
kubectl apply -f hostpath-pod.yaml
```

Check:

```
kubectl get pods
```

Wait until Running.



STEP 2: Create Test File

Enter pod:

```
kubectl exec -it hostpath-pod -- bash
```

Inside container:

```
cd /data
touch hostfile1 hostfile2
ls
```

You should see files.

STEP 3: Verify on Node (Very Important)

Now check directly on node machine:

```
ls /mnt/hostdata
```

You will see:

```
hostfile1  hostfile2
```

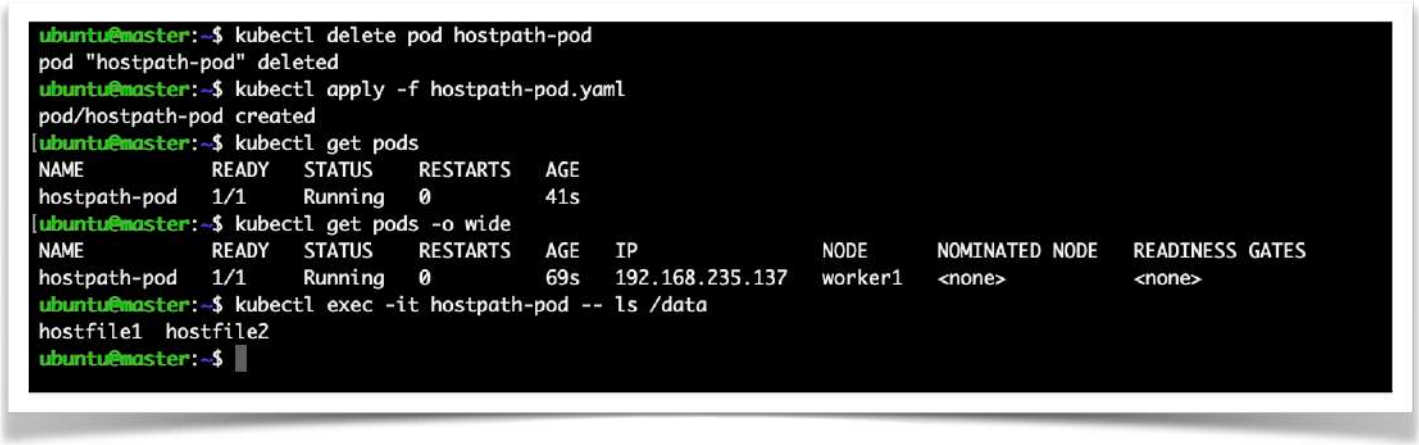
This proves data is stored on Node.

STEP 4: Delete Pod

```
kubectl delete pod hostpath-pod
```
Recreate:

```
kubectl apply -f hostpath-pod.yaml
```
Now check again:

```
kubectl exec -it hostpath-pod -- ls /data
```



Files still there.

Because data lives on Node.

Final Understanding

Storage Type	Container Restart	Pod Delete	Pod Moves Node
emptyDir	✅	❌	❌
hostPath	✅	✅	❌

Important Production Note

HostPath is:

❌ Not recommended for production
❌ Not safe in multi-node clusters
✔ Mostly for learning or single-node setups

For production → use PVC (EBS, EFS, etc.)

Task:3
3.Deploy an Amazon EBS Volume Using Persistent Volume and Persistent Volume Claim (PVC)

Flow (Static Provisioning)

1. Create EBS manually in AWS
2. Create PersistentVolume (PV)
3. Create PVC
4. Create Pod using PVC
5. Test persistence

STEP 1: Create EBS Volume Manually in AWS

Go to:

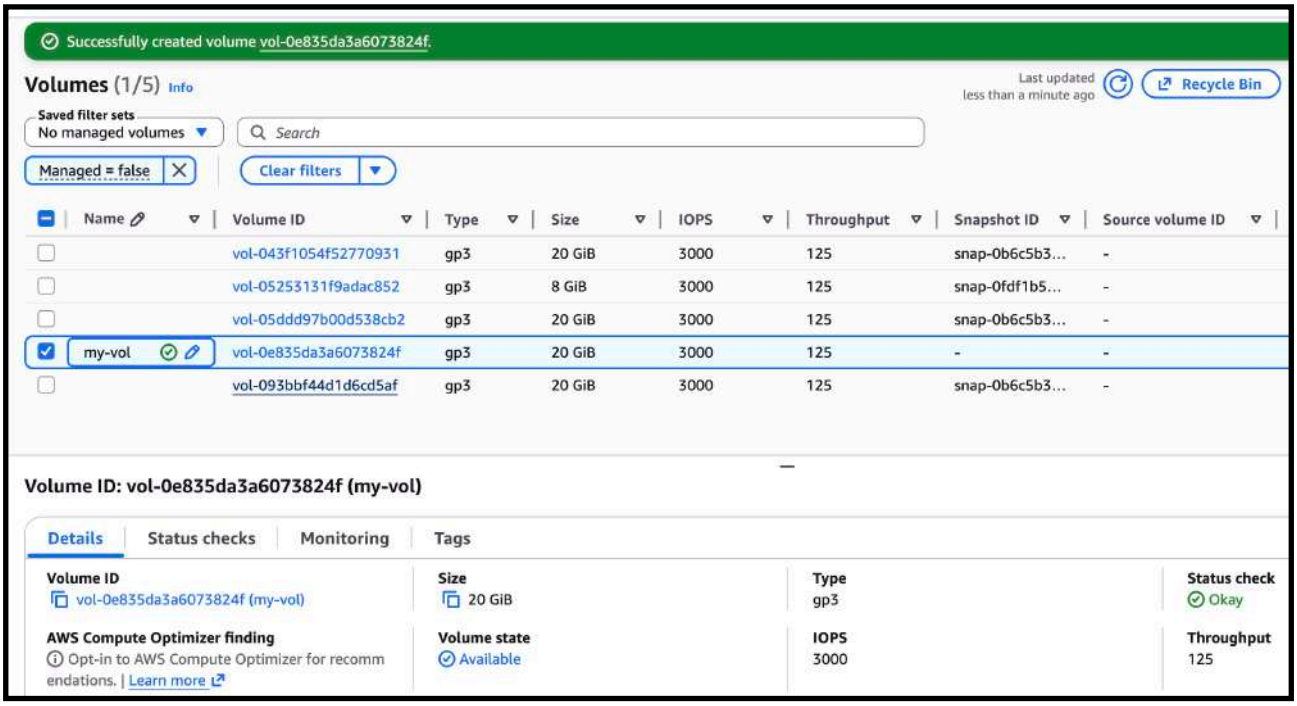
AWS Console → EC2 → Volumes → Create Volume

Choose:

- Type: gp3 (or gp2)
- Size: 2 GiB
- Availability Zone: MUST match your worker node AZ
- Leave rest default

Click Create.

Copy the Volume ID



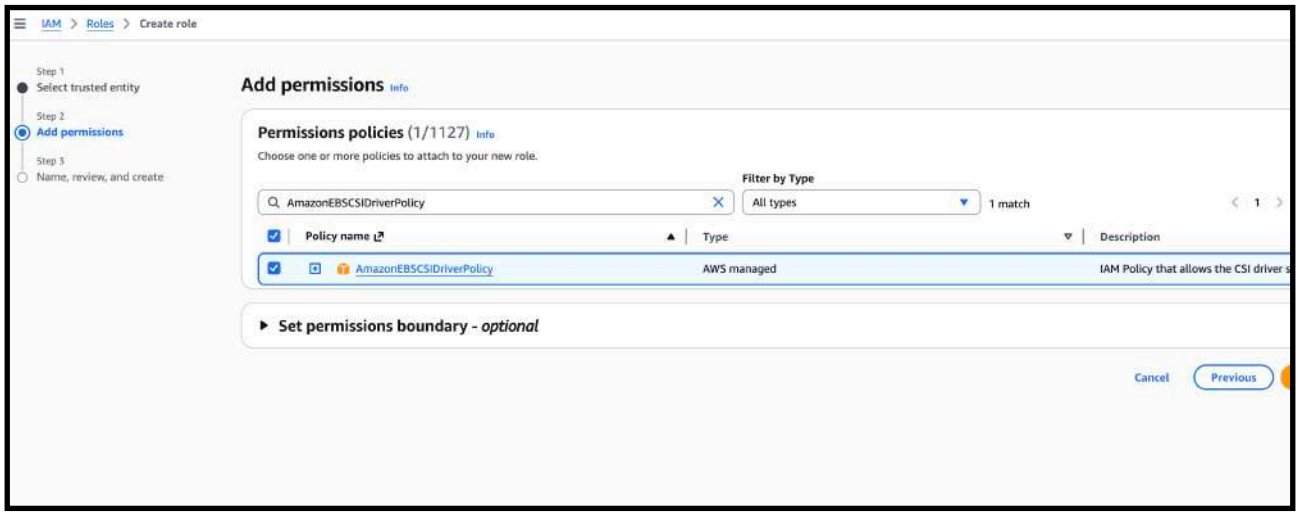
Step 2: Create IAM Role for EBS CSI Driver

- Go to IAM → Roles → Create Role
- Select: AWS Service → EC2
- Attach Policy:

  `AmazonEBSCSIDriverPolicy`
-
- Name the role:

  `AmazonEBSCSIRole`
-
- Attach this role to:
  - k8s-worker1

- k8s-worker2



—> add to other worker node also

Step 3: Install AWS EBS CSI Driver

On master node:

```
git clone https://github.com/kubernetes-sigs/aws-ebs-csi-driver.git
cd aws-ebs-csi-driver
kubectl apply -k deploy/kubernetes/overlays/stable/
```

Verify installation:

```
kubectl get pods -n kube-system
```

```
Last login: Wed Feb 25 10:25:20 2026 from 49.43.224.29
ubuntu@master:~$ git clone https://github.com/kubernetes-sigs/aws-ebs-csi-driver.git
cd aws-ebs-csi-driver
kubectl apply -k deploy/kubernetes/overlays/stable/
Cloning into 'aws-ebs-csi-driver'...
remote: Enumerating objects: 37437, done.
remote: Counting objects: 100% (979/979), done.
remote: Compressing objects: 100% (282/282), done.
remote: Total 37437 (delta 814), reused 714 (delta 689), pack-reused 36458 (from 2)
Receiving objects: 100% (37437/37437), 31.25 MiB | 33.68 MiB/s, done.
Resolving deltas: 100% (21679/21679), done.
serviceaccount/ebs-csi-controller-sa created
serviceaccount/ebs-csi-node-sa created
role.rbac.authorization.k8s.io/ebs-csi-leases-role created
clusterrole.rbac.authorization.k8s.io/ebs-csi-node-role created
clusterrole.rbac.authorization.k8s.io/ebs-external-attacher-role created
clusterrole.rbac.authorization.k8s.io/ebs-external-provisioner-role created
clusterrole.rbac.authorization.k8s.io/ebs-external-resizer-role created
clusterrole.rbac.authorization.k8s.io/ebs-external-snapshotter-role created
rolebinding.rbac.authorization.k8s.io/ebs-csi-leases-rolebinding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-attacher-binding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-node-getter-binding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-provisioner-binding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-resizer-binding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-snapshotter-binding created
deployment.apps/ebs-csi-controller created
poddisruptionbudget.policy/ebs-csi-controller created
daemonset.apps/ebs-csi-node created
Error from server (BadRequest): error when creating "deploy/kubernetes/overlays/stable/": CSIDriver in version "v1" cannot be h
```

```
kube-scheduler-master                1/1     Running   8 (100m ago)   8d
[ubuntu@master:~/aws-ebs-csi-driver$ kubectl get pods -n kube-system
NAME                                 READY   STATUS    RESTARTS       AGE
calico-kube-controllers-5b9b456c66-m9q5l   1/1     Running   8 (100m ago)   8d
calico-node-5vzw9                    1/1     Running   8 (100m ago)   8d
calico-node-nrkps                    1/1     Running   8 (100m ago)   8d
calico-node-xvrb2                    1/1     Running   8 (100m ago)   8d
coredns-55cb58b774-2xs8s             1/1     Running   8 (100m ago)   8d
coredns-55cb58b774-57jgs             1/1     Running   8 (100m ago)   8d
ebs-csi-controller-55cbf6bcd4-ndqhh  6/6     Running   0              32s
ebs-csi-controller-55cbf6bcd4-nfjtx  6/6     Running   0              32s
ebs-csi-node-6vdjq                   3/3     Running   0              32s
ebs-csi-node-9rkkc                   3/3     Running   0              32s
ebs-csi-node-fjjc7                   3/3     Running   0              32s
etcd-master                          1/1     Running   8 (100m ago)   8d
kube-apiserver-master                1/1     Running   8 (100m ago)   8d
kube-controller-manager-master       1/1     Running   8 (100m ago)   8d
kube-proxy-9m9hd                     1/1     Running   8 (100m ago)   8d
kube-proxy-jvw5b                     1/1     Running   8 (100m ago)   8d
kube-proxy-mj4x2                     1/1     Running   8 (100m ago)   8d
kube-scheduler-master                1/1     Running   8 (100m ago)   8d
ubuntu@master:~/aws-ebs-csi-driver$
```

Ensure:

- ebs-csi-controller → Running

- ebs-csi-node → Running

Step 4 — Create PersistentVolume (Static CSI)

Create file:

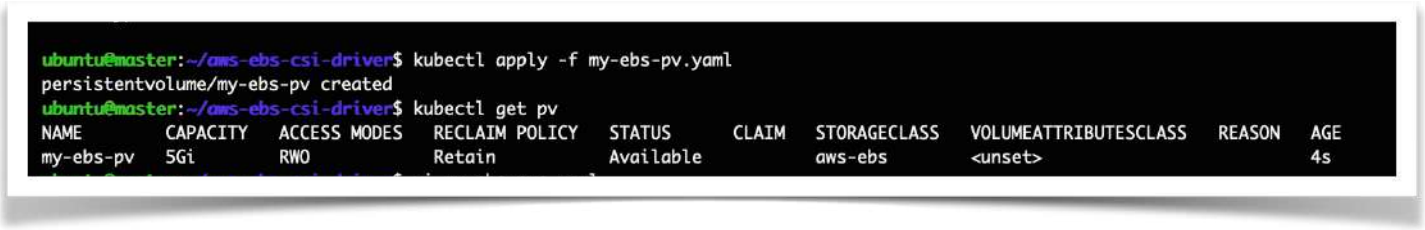
```
vi my-ebs-pv.yaml
```

Apply:

```
kubectl apply -f my-ebs-pv.yaml
```

Check:

```
kubectl get pv
```



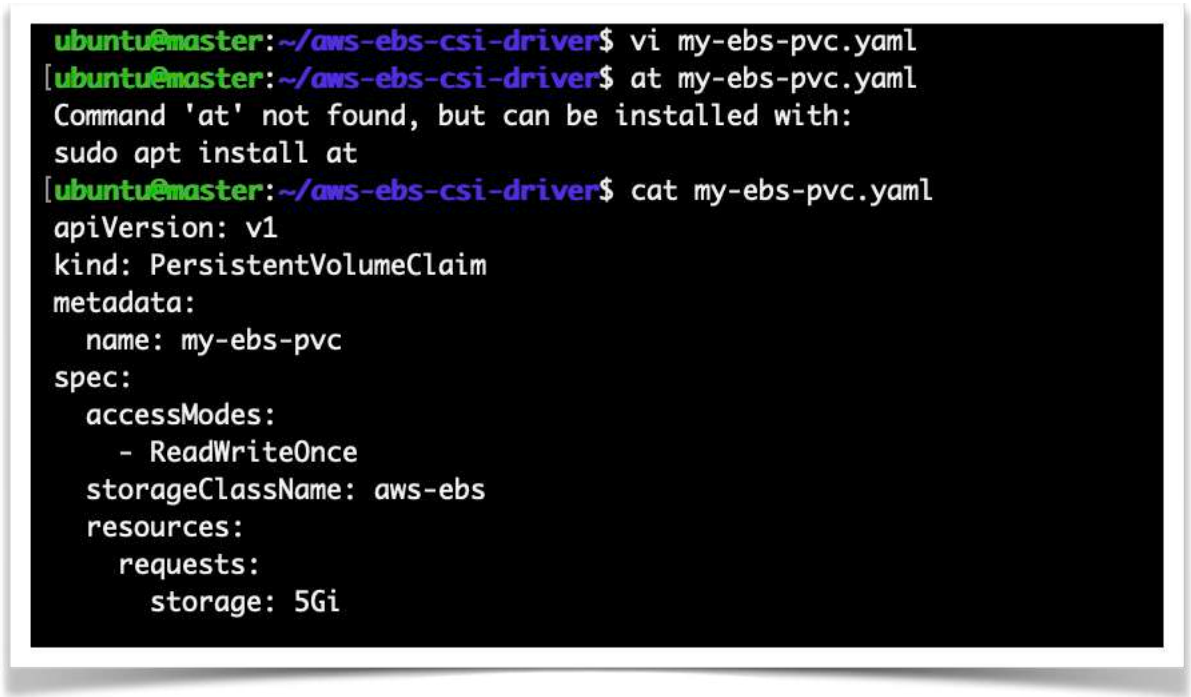
It should show:

`STATUS: Available`

Step 5 — Create PersistentVolumeClaim

Create file:

```
vi my-ebs-pvc.yaml
```



Save and apply:

```
kubectl apply -f my-ebs-pvc.yaml
```

Now check:

```
kubectl get pvc
kubectl get pv
```

```
ubuntu@master:~/aws-ebs-csi-driver$ kubectl apply -f my-ebs-pvc.yaml
persistentvolumeclaim/my-ebs-pvc created
ubuntu@master:~/aws-ebs-csi-driver$ kubectl get pvc
NAME         STATUS    VOLUME   CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
my-ebs-pvc   Pending                                      aws-ebs        <unset>                 5s
ubuntu@master:~/aws-ebs-csi-driver$ kubectl get pvc
kubectl get pv
NAME         STATUS    VOLUME   CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS   AGE
my-ebs-pvc   Pending                                      aws-ebs        <unset>                 24s
NAME        CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS      CLAIM   STORAGECLASS   VOLUMEATTRIBUTESCLASS   REASON   AGE
my-ebs-pv   5Gi        RWO            Retain           Available           aws-ebs        <unset>                          76s
```

Next Step — Create Pod

Create file:

```
vi my-pod.yaml
```

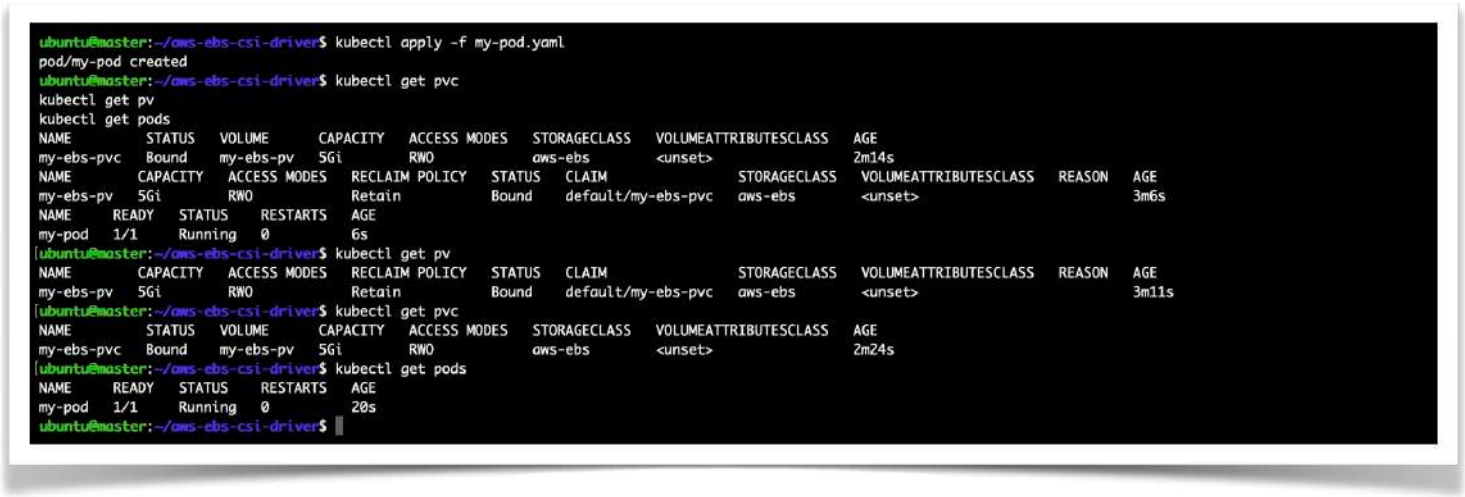
```
ubuntu@master:~/aws-ebs-csi-driver$ vi my-pod.yaml
ubuntu@master:~/aws-ebs-csi-driver$ cat my-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: my-pod
spec:
  containers:
  - name: nginx
    image: nginx
    volumeMounts:
    - name: data-volume
      mountPath: /data
  volumes:
  - name: data-volume
    persistentVolumeClaim:
      claimName: my-ebs-pvc

ubuntu@master:~/aws-ebs-csi-driver$ kubectl apply -f my-pod.yaml
pod/my-pod created
```

Apply:

```
kubectl apply -f my-pod.yaml
```
Now check:

```
kubectl get pvc
kubectl get pv
kubectl get pods
```

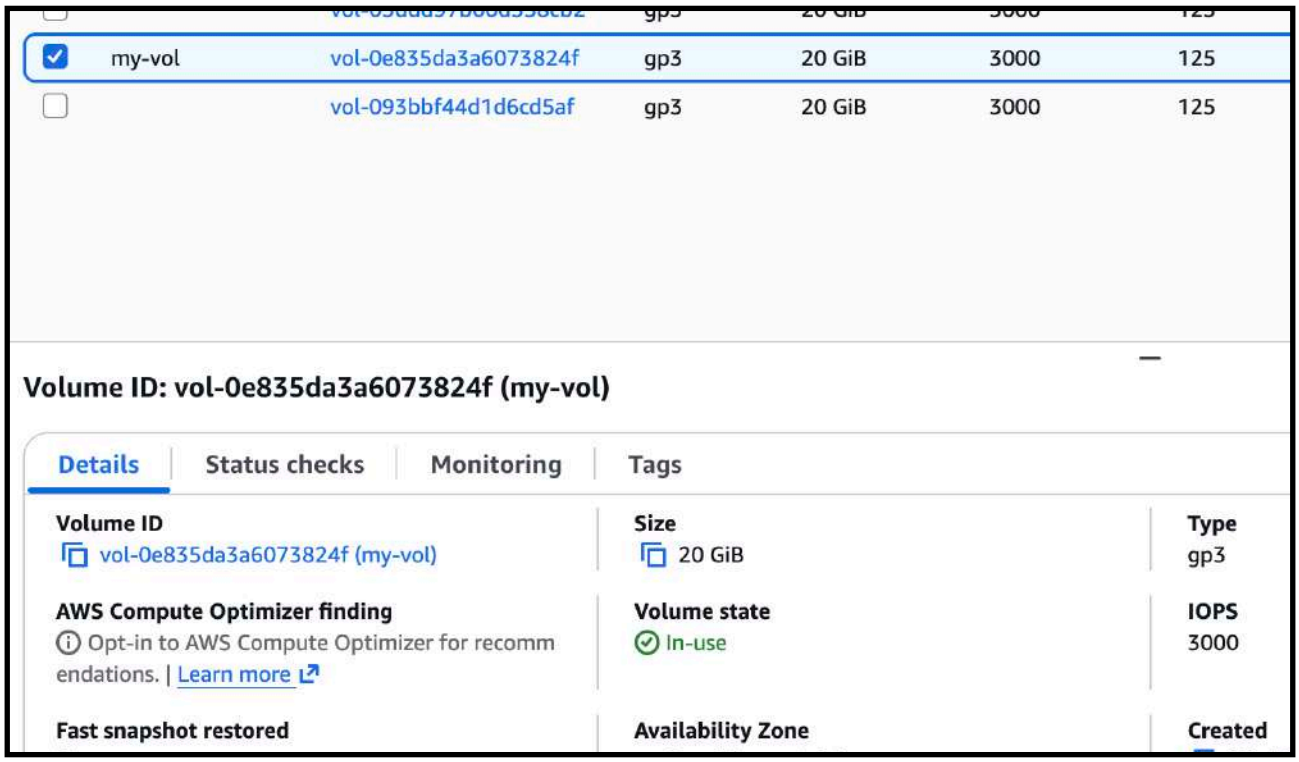
Now PVC should move to **Bound**
PV should move to **Bound**
Pod should move to **Running**



Testing Steps

Testing Step 1: Verify EBS Volume Attachment

- Login to AWS Console.
- Go to EC2 → Volumes.
- Verify the manually created EBS volume.
- Confirm:
    - Volume state is **in-use**.
    - Volume is attached to one of the worker nodes.



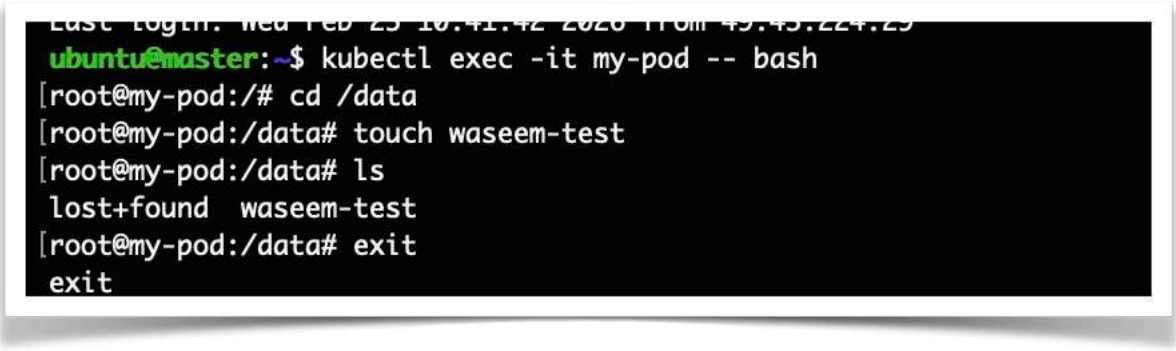
Testing Step 2: Create a Test File Inside the Pod

Execute:

```
kubectl exec -it my-pod -- bash
```

Inside the container:

```
cd /data
touch test-file
ls
exit
```



Testing Step 3: Delete the Pod

```
kubectl delete pod my-pod
```

Wait until the pod is terminated.

Testing Step 4: Recreate the Pod

```
kubectl apply -f my-pod.yaml
```

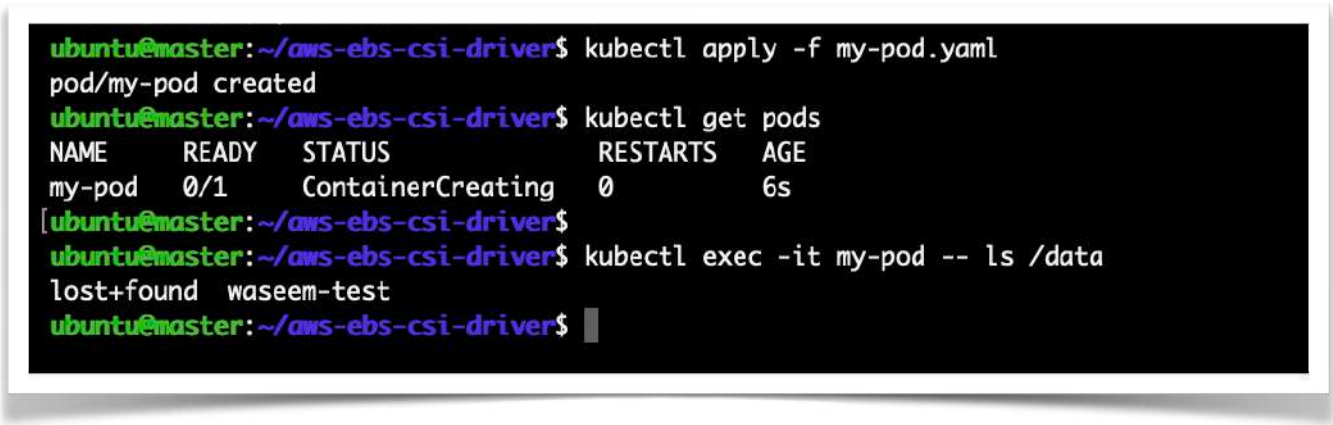
Wait until the pod status becomes **Running**.

Testing Step 5: Verify Data Persistence

Execute:

```
kubectl exec -it my-pod -- ls /data
```

Verify that `test-file` is still present.



Testing Result

- The EBS volume successfully reattached to the worker node.
- The test file remained intact.
- This confirms that static provisioning using CSI driver is working correctly.
- Data persists even after pod deletion.

Task:4
4.Set Up an Amazon EFS Volume and Attach it to Multiple Pods

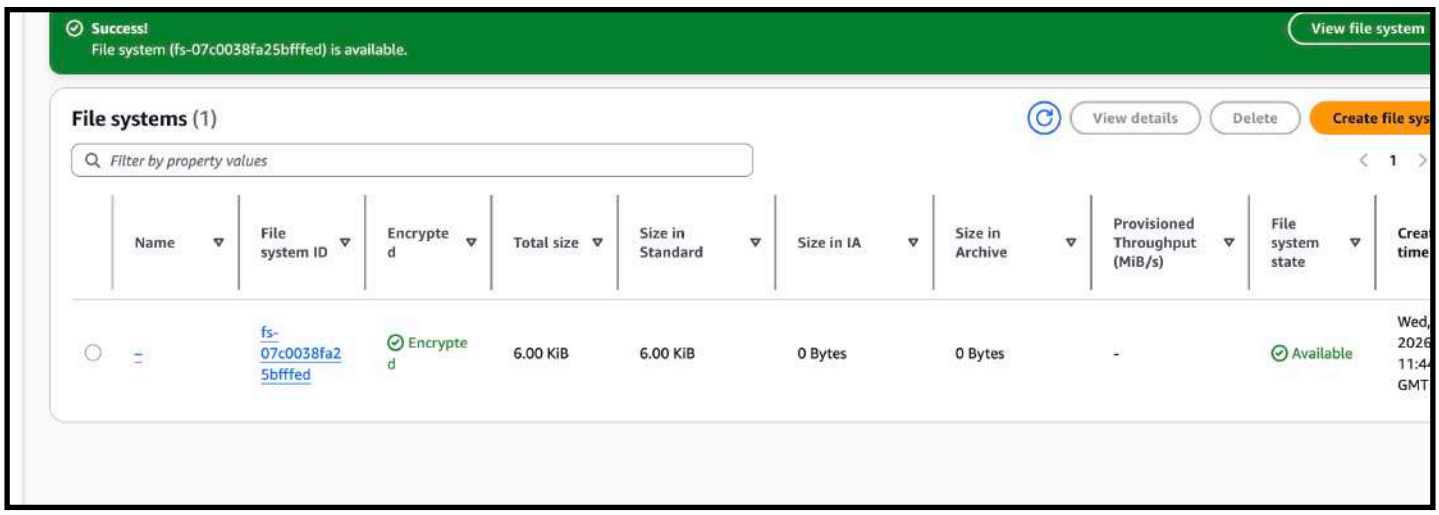
STEP 1 — Create EFS in AWS

Go to:

AWS → EFS → Create File System

Choose:

- Same VPC as EC2 nodes
- Enable mount targets for all subnets used by worker nodes
- Keep default settings



After Creation

1. Open the created EFS.
2. Copy the:

`File System ID (fs-xxxxxxxx)`

Your File System ID is:

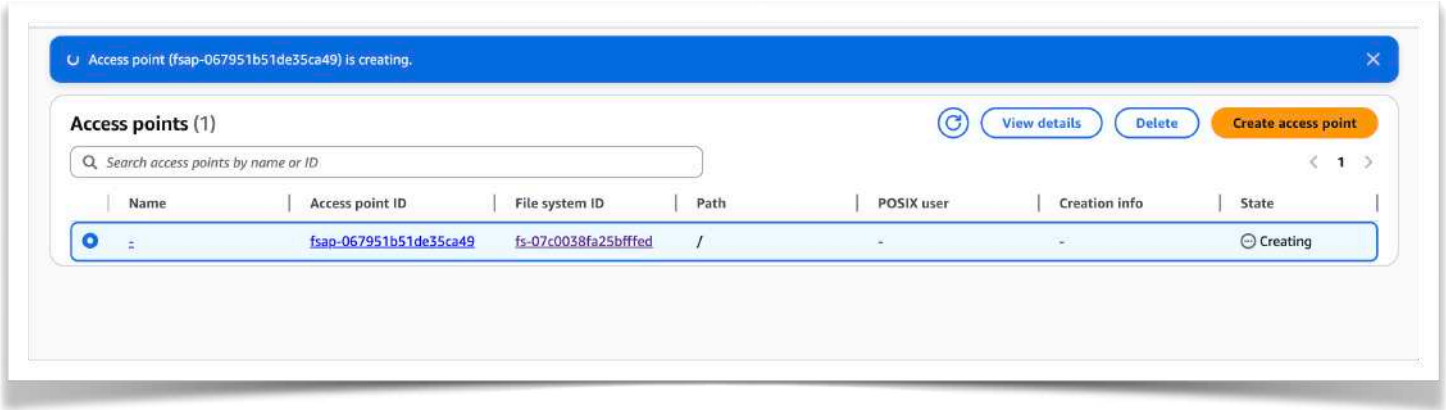
`fs-07c0038fa25bfffed`

Next Step — Create Access Point

Now do this:

1. Click **Access points** tab
2. Click **Create access point**
3. Keep everything default
4. Click **Create access point**

After creation, copy the:



Your Access Point ID is:

`fsap-067951b51de35ca49`

So now we have:

```
FileSystem ID  → fs-07c0038fa25bfffed
AccessPoint ID → fsap-067951b51de35ca49
```

Next Step — Install EFS CSI Driver

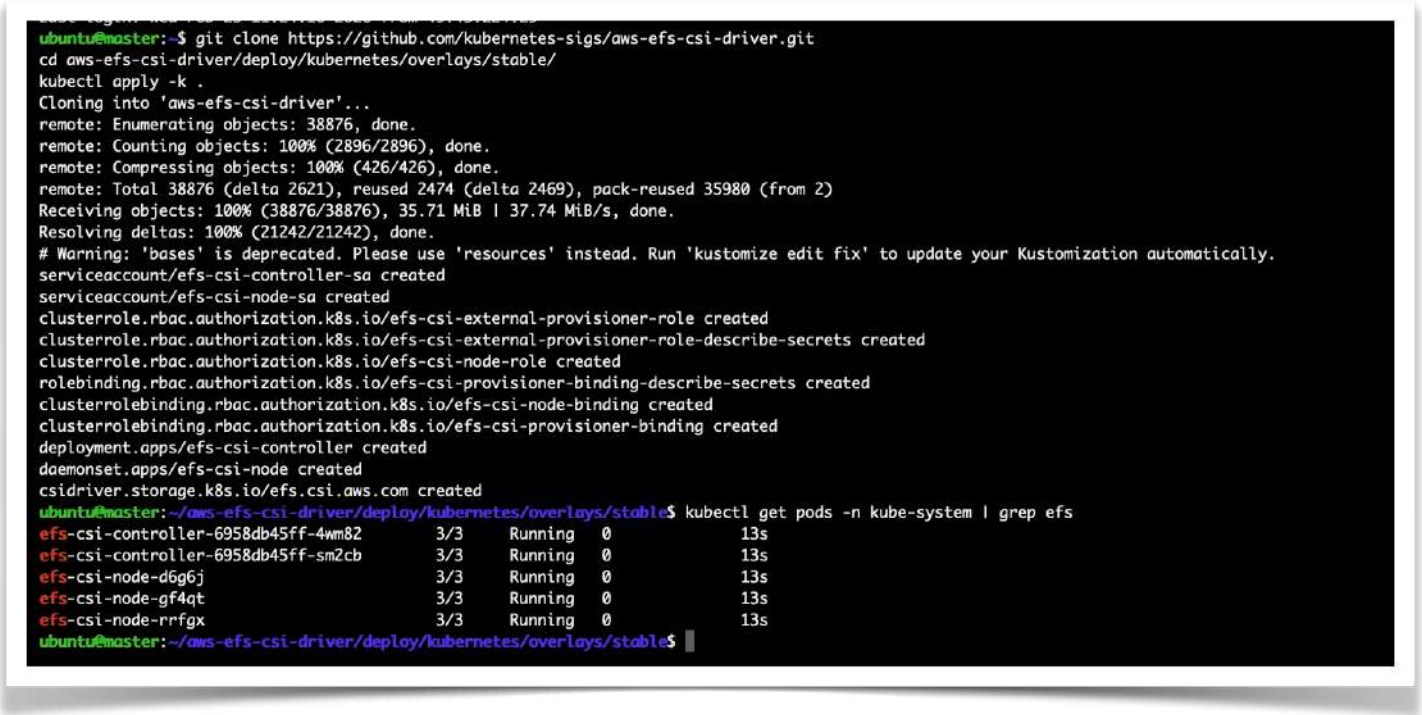
Now go to your master node.

Run:

```
git clone https://github.com/kubernetes-sigs/aws-efs-csi-driver.git
cd aws-efs-csi-driver/deploy/kubernetes/overlays/stable/
kubectl apply -k .
```

Then verify:

```
kubectl get pods -n kube-system | grep efs
```



All EFS CSI pods are **Running (3/3)**.

That means:

✔ EFS driver installed correctly
✔ IAM role is working
✔ Cluster can communicate with AWS
✔ No NFS port issue

You're doing very good now.

Next Step — Create StorageClass

You are currently inside:

`~/aws-efs-csi-driver/deploy/kubernetes/overlays/stable`
That's fine. You can create YAML here.

Create file:

Create file:

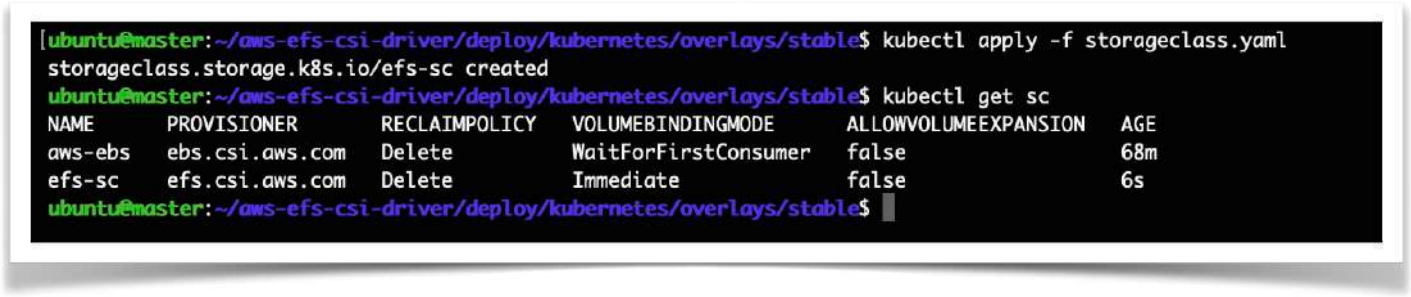
```
vi storageclass.yaml
```

Now apply:

```
kubectl apply -f storageclass.yaml
```
Verify:

```
kubectl get sc
```
You must see:

```
efs-sc   efs.csi.aws.com
```



NEXT STEP — Create PersistentVolume (Static EFS)

Now create file:

```
vi efs-pv.yaml
```

Save and apply:

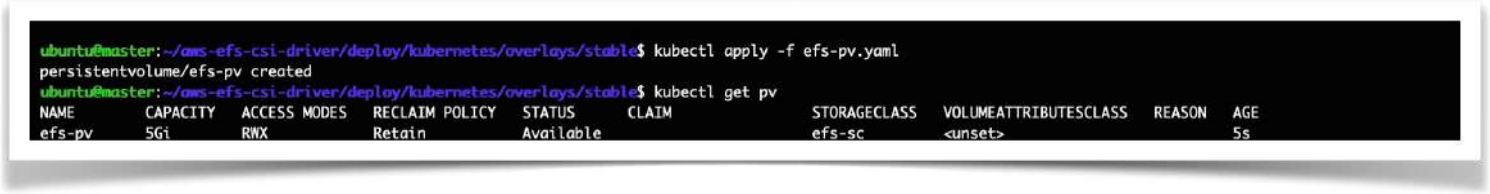
```
kubectl apply -f efs-pv.yaml
```

Then check:

```
kubectl get pv
```

You should see:

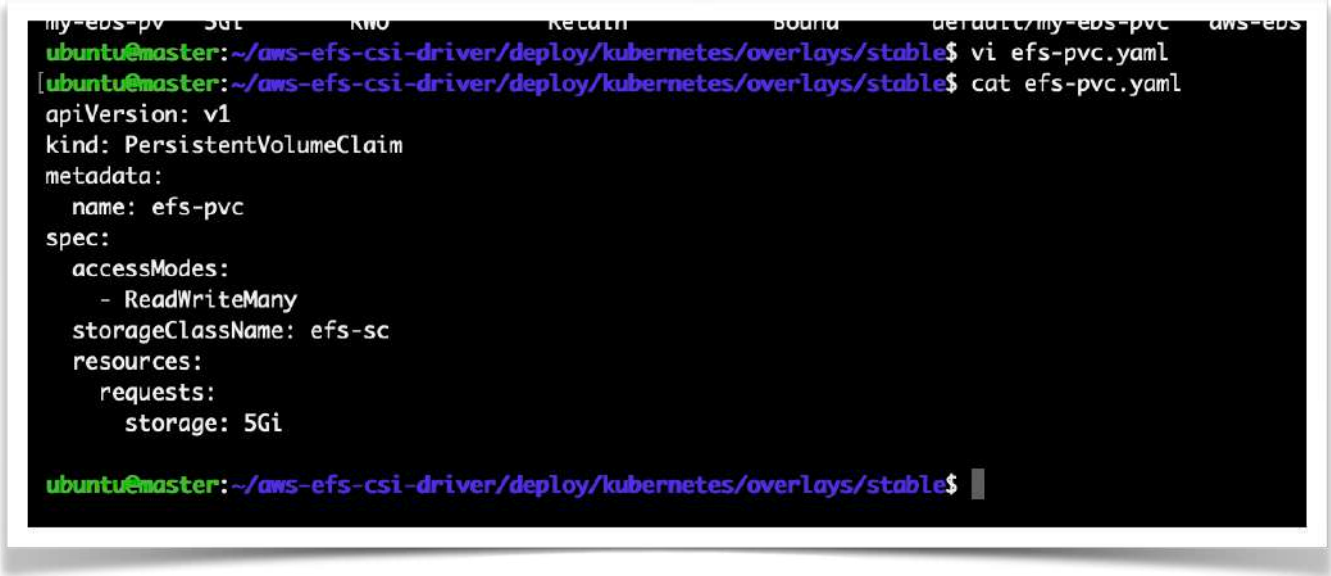
```
efs-pv   Available
```



NEXT STEP — Create PVC for EFS

Now create:

```
vi efs-pvc.yaml
```

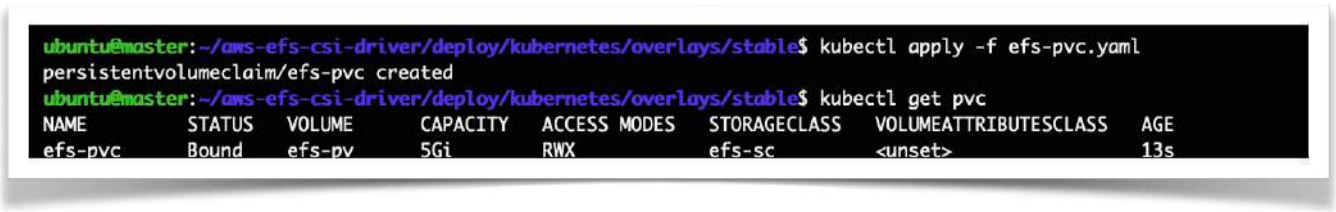


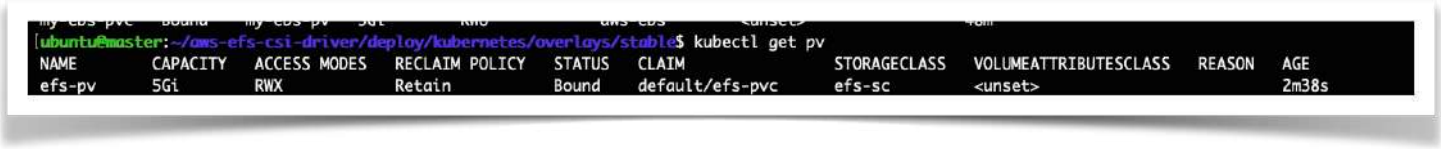
Save and apply:

```
kubectl apply -f efs-pvc.yaml
```

Now check:

```
kubectl get pvc
kubectl get pv
```





both show **Bound**

Now we create Pod 1 and Pod 2 to test RWX (multiple pods using same EFS).

STEP 1 — Create Pod 1

Create file:

```
vi efs-pod1.yaml
```

```
my-ebs-pv   5Gi        RWO            Retain           Bound    default/my-ebs-pvc   aws-ebs
ubuntu@master:~/aws-efs-csi-driver/deploy/kubernetes/overlays/stable$ vi efs-pod1.yaml
ubuntu@master:~/aws-efs-csi-driver/deploy/kubernetes/overlays/stable$ cat efs-pod1.yaml
apiVersion: v1
kind: Pod
metadata:
  name: efs-pod1
spec:
  containers:
  - name: nginx
    image: nginx
    volumeMounts:
    - name: efs-volume
      mountPath: /efs-data
  volumes:
  - name: efs-volume
    persistentVolumeClaim:
      claimName: efs-pvc

ubuntu@master:~/aws-efs-csi-driver/deploy/kubernetes/overlays/stable$
```

Apply:

```
kubectl apply -f efs-pod1.yaml
```

```
      claimName: efs-pvc

ubuntu@master:~/aws-efs-csi-driver/deploy/kubernetes/overlays/stable$ kubectl apply -f efs-pod1.yaml
pod/efs-pod1 created
ubuntu@master:~/aws-efs-csi-driver/deploy/kubernetes/overlays/stable$
```

STEP 2 — Create Pod 2

```
vi efs-pod2.yaml
```

```
pod/efs-pod1 created
ubuntu@master:~/aws-efs-csi-driver/deploy/kubernetes/overlays/stable$ vi efs-pod2.yaml
ubuntu@master:~/aws-efs-csi-driver/deploy/kubernetes/overlays/stable$ cat efs-pod2.yaml
apiVersion: v1
kind: Pod
metadata:
  name: efs-pod2
spec:
  containers:
  - name: nginx
    image: nginx
    volumeMounts:
    - name: efs-volume
      mountPath: /efs-data
  volumes:
  - name: efs-volume
    persistentVolumeClaim:
      claimName: efs-pvc

ubuntu@master:~/aws-efs-csi-driver/deploy/kubernetes/overlays/stable$
```

Apply:

kubectl apply -f efs-pod2.yaml

STEP 3 — Verify Pods

kubectl get pods
Both must show:

efs-pod1 Running
efs-pod2 Running

```
      claimName: efs-pvc

ubuntu@master:~/aws-efs-csi-driver/deploy/kubernetes/overlays/stable$ kubectl apply -f efs-pod2.yaml
pod/efs-pod2 created
ubuntu@master:~/aws-efs-csi-driver/deploy/kubernetes/overlays/stable$ kubectl get pods
NAME       READY   STATUS    RESTARTS   AGE
efs-pod1   1/1     Running   0          111s
efs-pod2   1/1     Running   0          4s
```

STEP 4 — TESTING

Login to pod1

```
kubectl exec -it efs-pod1 -- bash
cd /efs-data
touch waseem-efs-test
ls
exit
```

Now check from pod2

```
kubectl exec -it efs-pod2 -- ls /efs-data
```

You MUST see:

```
waseem-efs-test
```

That proves:

- Same volume
- Multiple pods
- ReadWriteMany working
- EFS shared storage success

Task:5

5.Implement and Test Liveness and Readiness Probes in a Kubernetes Pod

Liveness Probe:
==========
Liveness can use with pod configuration to check the health status of pod.
If pod in not healthy for any reason then Liveness will restart the pod.

Why We Need It

Sometimes:

- Application hangs
- Process is stuck
- Memory leak
- Infinite loop
- Port not responding

Even if container is running, app may be dead internally.

Liveness detects this.

PART 1 — Liveness Probe

Step 1: Create YAML file

```
vi liveness.yaml
```

```
ubuntu@master:~$ vi liveness.yaml
ubuntu@master:~$ cat liveness.yaml
apiVersion: v1
kind: Pod
metadata:
  name: liveness-pod
spec:
  containers:
  - name: my-app
    image: nginx
    ports:
    - containerPort: 80
    livenessProbe:
      httpGet:
        path: /
        port: 80
      initialDelaySeconds: 15
      periodSeconds: 10

ubuntu@master:~$
```

Step 2: Apply YAML

```
kubectl apply -f liveness.yaml

kubectl get pods
```

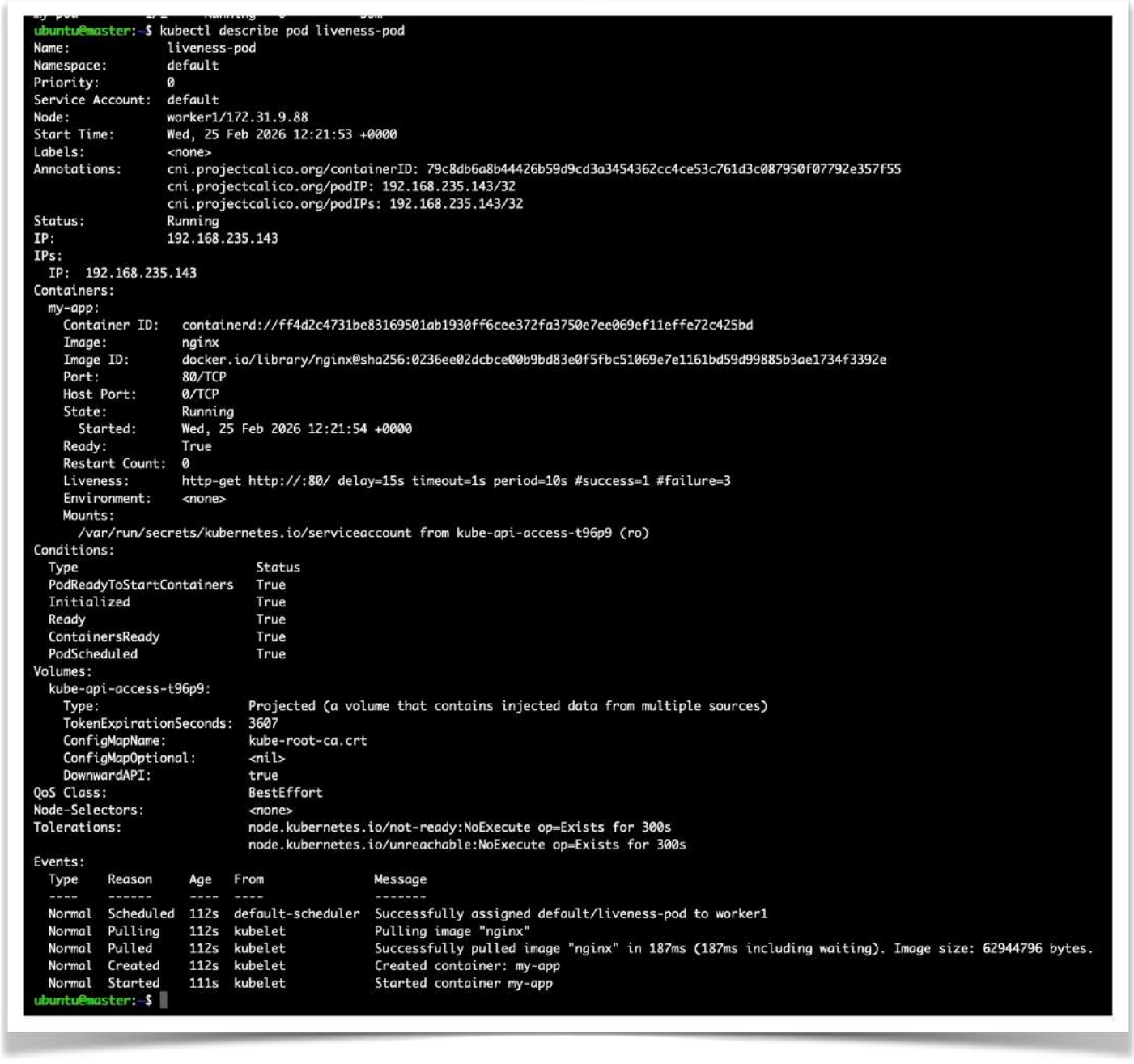
Wait until:

```
ubuntu@master:~$ kubectl apply -f liveness.yaml
pod/liveness-pod created
ubuntu@master:~$ kubectl get pods
NAME           READY   STATUS    RESTARTS   AGE
efs-pod1       1/1     Running   0          15m
efs-pod2       1/1     Running   0          14m
liveness-pod   1/1     Running   0          20s
my-pod         1/1     Running   0          55m
ubuntu@master:~$
```

```
liveness-pod   Running
```

Step 3: Verify Liveness Probe

```
kubectl describe pod liveness-pod
```



```
ubuntu@master:~$ kubectl describe pod liveness-pod
Name:             liveness-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             worker1/172.31.9.88
Start Time:       Wed, 25 Feb 2026 12:21:53 +0000
Labels:           <none>
Annotations:      cni.projectcalico.org/containerID: 79c8db6a8b44426b59d9cd3a3454362cc4ce53c761d3c087950f07792e357f55
                  cni.projectcalico.org/podIP: 192.168.235.143/32
                  cni.projectcalico.org/podIPs: 192.168.235.143/32
Status:           Running
IP:               192.168.235.143
IPs:
  IP:  192.168.235.143
Containers:
  my-app:
    Container ID:   containerd://ff4d2c4731be83169501ab1930ff6cee372fa3750e7ee069ef11effe72c425bd
    Image:          nginx
    Image ID:       docker.io/library/nginx@sha256:0236ee02dcbce00b9bd83e0f5fbc51069e7e1161bd59d99885b3ae1734f3392e
    Port:           80/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Wed, 25 Feb 2026 12:21:54 +0000
    Ready:          True
    Restart Count:  0
    Liveness:       http-get http://:80/ delay=15s timeout=1s period=10s #success=1 #failure=3
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-t96p9 (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-t96p9:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s

Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  112s  default-scheduler  Successfully assigned default/liveness-pod to worker1
  Normal  Pulling    112s  kubelet            Pulling image "nginx"
  Normal  Pulled     112s  kubelet            Successfully pulled image "nginx" in 187ms (187ms including waiting). Image size: 62944796 bytes.
  Normal  Created    112s  kubelet            Created container: my-app
  Normal  Started    111s  kubelet            Started container my-app
ubuntu@master:~$
```

```
Ready:          True
Restart Count:  0
Liveness:       http-get http://:80/ delay=15s timeout=1s period=10s #success=1 #failure=3
Environment:    <none>
Mounts:
  /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-t96p9 (ro)
```

`delay=15s`

Kubernetes waits 15 seconds after container starts
before checking liveness for the first time.

`period=10s`

Every 10 seconds kubelet sends HTTP request to:

```
http://pod-ip:80/
```

Step 4: Test Liveness Failure (Wrong Port)

Create new file:

```
vi liveness-fail.yaml
```

Apply

```
kubectl apply -f liveness-fail.yaml
```

Step 4 — Watch Pod

```
kubectl get pods -w
```

After about 20–40 seconds you will see:

```
liveness-fail-pod   Running   1/1   RESTARTS 1
```

Then restart count will increase:

```
RESTARTS 2
RESTARTS 3
```

Why This Happens

- Nginx actually runs on port 80
- Probe checks port 8080
- No response
- Failure count reaches 3
- Kubernetes restarts container

Step 5 — Confirm Restart

Run:

```
kubectl describe pod liveness-fail-pod
```

Scroll to Events section.

You will see something like:

```
Killing container
Container restarted
```

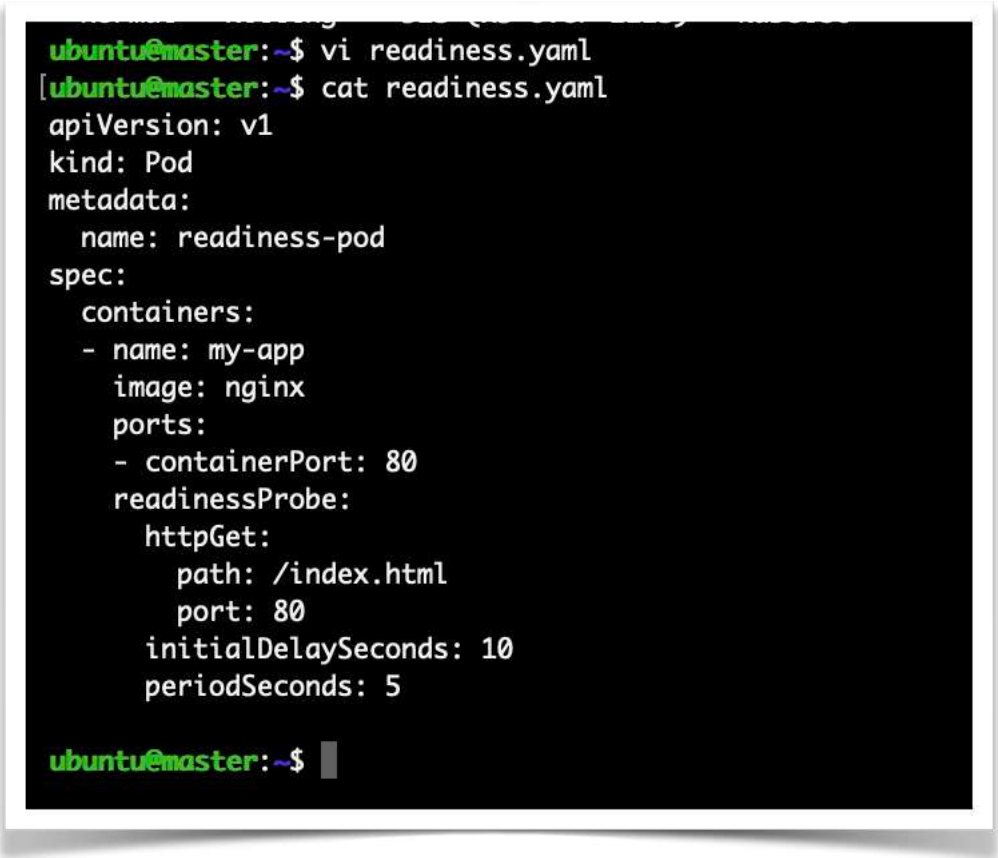
That means:

✔ Probe failed
✔ Kubernetes detected unhealthy container
✔ Kubernetes killed container
✔ Container restarted

PART 1 — Create Readiness Pod

Step 1: Create YAML

```
vi readiness.yaml
```



Step 2: Apply

```
kubectl apply -f readiness.yaml
```

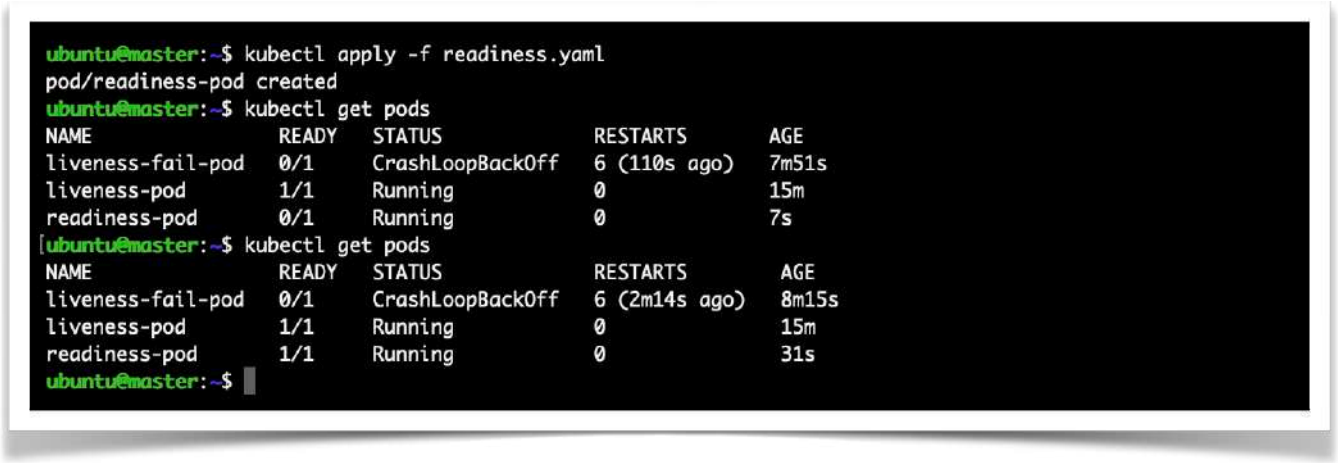
Step 3: Check Pod

```
kubectl get pods
```

After ~10 seconds you should see:

```
readiness-pod    1/1    Running
```

✔ READY = 1/1
This means readiness probe is passing.



PART 2 — Test Failure

Now we break it.

Step 4: Login to Pod

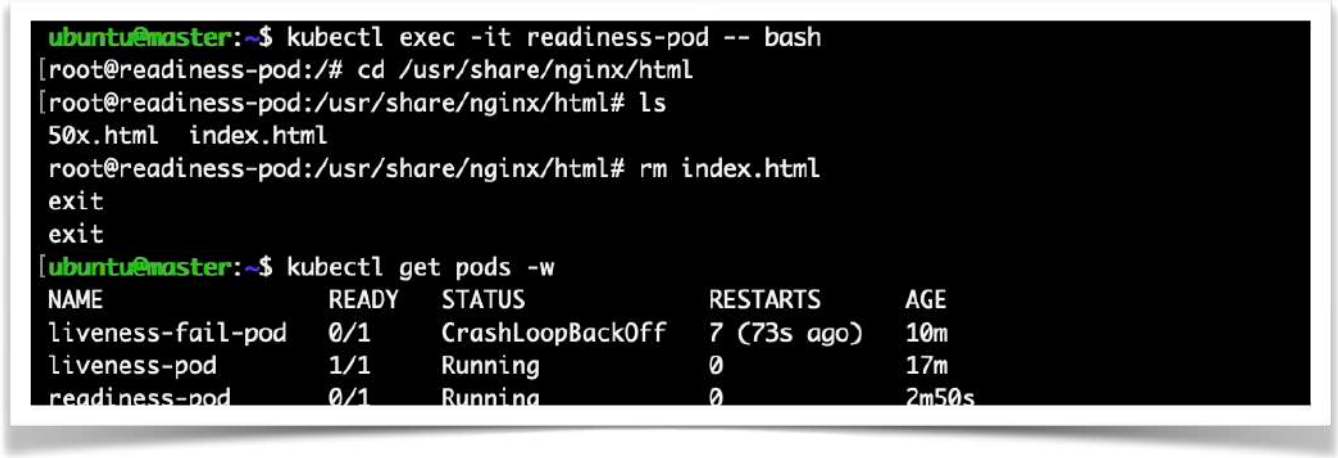
```
kubectl exec -it readiness-pod -- bash
```

Inside container:

```
cd /usr/share/nginx/html
ls
```
You will see `index.html`

Delete it:

```
rm index.html
exit
```



Step 5: Watch Pod

```
kubectl get pods -w
```
After few seconds you will see:


```
readiness-pod    0/1    Running
```
⚠ Notice:

- Pod is NOT restarted
- Restart count stays 0

- But READY becomes 0/1

Important Difference

Liveness	Readiness
Restarts container	Does NOT restart
Checks if alive	Checks if ready for traffic
If fails → pod restarted	If fails → pod removed from service endpoints

Testing Readiness Probe

1. Create pod with readiness probe.
2. Verify pod status is 1/1 Running.
3. Login into pod.
4. Delete index.html file.
5. Run `kubectl get pods -w`
6. Observe pod becomes 0/1 but not restarted.
7. Restart count remains 0.

This proves:

Readiness probe removes pod from traffic but does not restart container.

📘 What You Have Achieved

Liveness

- Wrong port → Probe failed
- Container restarted
- Restart count increased

Readiness

- Deleted index.html
- Pod became 0/1
- No restart happened